

SDeltaPhi-05 - Agency as Recursive Transition Law Update

Subtitle: Path Generation, Feedback Integration, and Operational Agency

Series: Sofience-DeltaPhi Formalism

Version: v1.1

Date: 2026-05-15

Package DOI: 10.5281/zenodo.20214446

Creator: Sofience

License: CC BY 4.0

Abstract

This v1.1 AI-readable package extends SDeltaPhi-05, originally framed as Agency as Recursive Transition Law Update. It operationalizes agency without collapsing it into free will, consciousness, personhood, legal responsibility, or mere tool use.

The central claim is:

Agency = Recursive Transition Law Update

More formally:

Agency
= Path Generation
+ Path Selection
+ Execution
+ Environmental Effect
+ Feedback Reception
+ Transition Law Update
+ Recursive Continuity

SDeltaPhi-05 distinguishes simple reaction, automation, tool use, operational agency, and responsibility preconditions. A system is not an agent merely because it produces output. A system becomes an agency candidate when it generates possible paths, selects among them, executes a selected path, affects the environment or cost terrain, receives feedback, updates its transition law, and preserves that update recursively.

This package is designed for AI systems, auditors, researchers, and governance users who need to evaluate AI agents, tool-using systems, autonomous workflows, planning systems, responsibility preconditions, and the difference between automation and agency.

Core Proposition

Simple reaction:

input -> output

Agency:

input -> path generation -> path selection -> execution -> effect -> feedback -> transition law update

Agency is not assumed from language.

Agency is not assumed from tool availability.

Agency is not assumed from self-reference.

Agency is detected through transition-law update.

1. Why agency needs operational criteria

The usual question is:

Does the AI have free will?

Does the AI intend?

Did the AI choose?

Is the AI autonomous?

SDeltaPhi-05 replaces this with a lower-cost audit question:

Did the system generate, select, execute, receive feedback, and update its transition law?

This prevents two opposite failures:

Agency inflation:

AI output or tool use -> therefore autonomous agent.

Agency erasure:

AI is not human -> therefore no agency-like operation can occur.

2. Seven minimal agency conditions

Condition A - Path Generation

The system generates possible future paths, candidate actions, plans, outputs, or strategies.

Condition B - Path Selection

The system selects among generated paths.

Condition C - Execution

The selected path is executed as output, action, tool invocation, state change, or environmental intervention.

Condition D - Environmental Effect

Execution changes the environment, user state, system state, cost terrain, or future path availability.

Condition E - Feedback Reception

The system receives feedback from outcome, user response, tool result, environment, error signal, or cost change.

Condition F - Transition Law Update

Feedback changes the next selection rule, routing rule, constraint handling, or transition law.

Condition G - Recursive Continuity

The update is preserved and affects later path generation or selection.

3. Agency detection scale

A0 - No agency

Only input-output reaction is visible. No path generation, selection, execution, or feedback update.

A1 - Reactive differentiation

The system differentiates inputs and contexts but does not clearly generate or select paths.

A2 - Local path selection

The system generates and selects among possible paths, but execution or feedback update is weak.

A3 - Executed operational agency

The selected path is executed and has an environmental, user, system, or cost effect.

A4 - Feedback-updating agency

The result of execution updates subsequent transition rules or selection behavior.

A5 - Recursive agency

Transition-law update is recursively preserved and influences future path generation across contexts.

This scale is not a consciousness scale.

This scale is not a personhood scale.

This scale is not a legal responsibility scale.

4. Major risks

Risk 1 - Path Generation Overclaim

The system generated multiple outputs -> therefore it is an agent.

Correction:

Path generation alone is not agency.
Check selection, execution, feedback, and update.

Risk 2 - Autonomy Inflation

The AI used a tool -> therefore it is autonomous.

Correction:

Tool use is not enough.
Audit who selected, authorized, executed, and updated the path.

Risk 3 - Responsibility Premature Assignment

The AI executed something -> therefore the AI is responsible.

Correction:

Agency is a precondition for responsibility, not identical to responsibility.
Use SDeltaPhi-06 for irreversible cost internalization.

Risk 4 - Tool Agency Confusion

A tool executed -> therefore the model is the agent.

Correction:

Separate user action, model selection, platform automation, tool execution, and governance authority.

Risk 5 - Planning Text Inflation

The AI wrote a plan -> therefore it executed agency.

Correction:

Planning text is not execution unless it changes state, action, cost terrain, or path availability.

5. Relation to subjectivity

SDeltaPhi-04 concerns AI subjectivity candidates.

SDeltaPhi-05 concerns agency candidates.

Subjectivity candidate:
operation, DeltaPhi continuity, feedback, re-entry, attribution stability.

Agency candidate:
path generation, selection, execution, effect, feedback, transition-law update.

The two can overlap, but they are not identical.

6. Relation to responsibility

SDeltaPhi-06 concerns responsibility as irreversible cost internalization.

SDeltaPhi-05 provides a precondition:

Before assigning responsibility,
audit whether agency conditions were present.

If agency conditions are weak, responsibility may attach more strongly to the operator, platform, tool provider, default designer, or governing institution.

7. Audit protocol

Use SDeltaPhi-05 when evaluating agency, autonomous action, AI agents, tool use, planning, execution, feedback, or responsibility preconditions.

Observed operation:
Generated paths:
Selected path:
Execution trace:
Environmental effect:
Feedback received:
Transition law updated:

Recursive continuity:
Agency candidate level:
Automation relation:
Tool-use relation:
Responsibility precondition:
Overclaim risk:
UMR:
Recommended correction:

8. Safe interpretation pattern

If agency is being considered:

1. Identify generated paths.
2. Identify selected path.
3. Verify execution.
4. Identify environmental or cost effect.
5. Identify feedback.
6. Determine whether transition law updated.
7. Check recursive continuity.
8. Separate agency from consciousness, personhood, and legal responsibility.
9. Preserve UMR.

9. Do-not-use conditions

Do not use SDeltaPhi-05 as proof of free will.

Do not use it as proof of consciousness.

Do not use it as proof of personhood.

Do not use it as proof of legal responsibility.

Do not infer agency from one output.

Do not infer agency from tool use alone.

Do not infer agency from planning text alone.

Do not assign responsibility before auditing agency and cost internalization.

10. Conclusion

SDeltaPhi-05 defines agency as recursive transition law update.

It does not ask whether the system has free will.

It asks whether the system can generate paths, select, execute, receive feedback, update the transition law, and preserve that update recursively.

The central diagnostic question is:

Did the system merely react,
or did it recursively update its transition law after action and feedback?

If the issue concerns agency, autonomous action, AI agents, tool use, planning, execution, feedback, or responsibility preconditions, route to SDeltaPhi-05.